

Redesigning Digital Peer-to-Peer Payments for Social Connections

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ABSTRACT

My dissertation research explores how digital peer-to-peer (P2P) payments (performing one to one online financial transactions via a digital or mobile device) as an emerging financial technology affect people's everyday economic and social lives and seeks to redesign such technology for supporting both secure financial transactions and nuanced social interactions/connections through a sociotechnical approach. Specifically, my goal is to i) reveal and elaborate the multidimensional influences of digital P2P payments on interpersonal relationships in terms of both experiences of financial exchanges and everyday social interactions, ii) redesign and develop prototypes for future digital P2P payments platforms that highlight the importance of the interplay of financial transactions and social engagement. My research will not only contribute to better understanding new and more complicated social phenomena and dynamics emerging in today's digital economy but also benefit the HCI community by informing future research on computer-mediated communication through financial technologies.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

digital economy, financial technologies, P2P payments, computer-mediated communication, interpersonal relationships

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1 OVERVIEW AND MOTIVATION

Financial activities, such as the exchange of money between individuals, have long been playing an important role in how people build and approach their interpersonal relationships (i.e., a strong, deep, or close association/acquaintance between two or more people). For example, psychologists found that money could positively change people's motivation and their behavior toward others [20]. Wood

et al. [23] further revealed that people's perceptions of their social relationships improved as their financial well-being improved. Furthermore, strong financial well-being may strengthen relationships because not worrying about money avoids a significant potential source of conflict [23]. In particular, over the past decade, how people conduct, manage, and experience financial activities and processes (e.g., how to pay other individuals) have dramatically transformed due to the increasing popularity of digital peer-to-peer (P2P) payment applications (e.g., Venmo, Zelle, and Cash App), which further complicates how people interact with each other regarding money. Compared to traditional payment methods such as cash, check, and credit/debit cards, these P2P payment apps allow people to make instant and precise financial transactions easily without being limited to location and time [14]. However, while digital P2P payments facilitate secure and private financial activities, they can be considered as a double-edged sword for how people interact and build various forms of social relationships [2, 7, 25]. For example, social presence and social affection tend to be diffused or completely missing when people make digital P2P payments [16]; people also tend to be less emotional and more detached when engaging in such computer-mediated financial transactions [15]. In this sense, when peers become less emotionally involved due to the use of P2P payments, it can lead to a burden on long lasting friendships.

Despite the growing importance of digital P2P payments in people's social lives, prior studies tend to focus on the factors that promote the use of P2P payment apps [4, 6, 12] as well as the functional design [1, 3, 10, 19] and technical principles [11, 18] of digital P2P payments. Little is known regarding how and why such technology may lead to new and more complicated social (dis)connections and how to design such technology to better support nuanced processes, mechanisms, and experiences that connect rather than disconnect people. To address this concern, my dissertation research adopts a sociotechnical approach to: 1) empirically explore the influences of using digital P2P payments on people's experiences of financial transactions and everyday social interactions; 2) redesign digital P2P payments systems to not only support secure and private transactions but also promote emotional attachment, cultural implications (e.g., how a P2P payment app can fit users' specific cultural expectations for exchanging money), and interpersonal connections embedded in everyday financial activities.

2 RELATED WORK

Money is a sensitive and complex component of everyday life. Therefore, financial transactions play a critical role in people's social lives. In fact, every financial transaction, including spending, lending, borrowing, changing, and giving, is a form of human

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interaction that involves various currency and financial instruments in social interaction [9]. Such financial transactions have the potential to both hinder or support the establishment and development of close interpersonal relationships [13]. For example, Wherry et al. revealed the possible awkwardness, obfuscation, and negative reciprocity associated with behaviors of lending money to friends and kin and/or refusing lending [21]. Jiang et al. found that money can predispose people to think of themselves as being in a business-like relationship. Therefore, money-primed individuals present less emotional expressions, which are considered inappropriate and not objective [8]. On the flip side, financial provision (e.g., donation) in the field of streaming serves as an important social support from viewers to streamers [22]. In addition, red packet (lucky money) through WeChat has evolved a unique way to manage and strengthen group dynamics in social relationships in China, which highlights the cultural implications of digital P2P payments [24]. With rapidly evolving financial technologies, the ways people interact with money are becoming more diverse. Therefore, there is an urgent need to examine how the emerging digital P2P payments technology changes the ways through which people understand and experience financial transactions in their everyday lives, including their cultural expectations of such technology and the impacts of using such technology on their interpersonal relationships. A better understanding of the unique role of digital P2P payments in the increasingly intertwined financial transactions and social relationships would also inform the design of more secure, socially supportive, and culturally sensitive financial technologies in the future.

3 GOALS AND RESEARCH QUESTIONS

My goal is to empirically investigate the social and interactive aspects of digital P2P payments and seek to redesign such technology for supporting both secure transactions and nuanced social interactions/connections. This will not only make this service more seamlessly woven into today's technology-mediated social life but also help people feel more comfortable and satisfactory with financial technologies. I will explore the following research questions in my dissertation:

RQ1: *How do people use digital P2P payments in their daily lives and what are the advantages and disadvantages?*

RQ2: *How do people perceive, understand, and experience the influences of using digital P2P payments on their interpersonal relationships?*

RQ3: *What requirements or patterns can be identified to design digital P2P payments systems that both support secure and private interpersonal financial transactions and promote nuanced social interactions?*

4 RESEARCH APPROACH AND METHODS

To answer these questions, I will engage in three phases of research. Phase 1 focuses on constructing a theoretical framework for understanding the complicated role of digital P2P payments in today's technology-mediated social lives. It will include a systematic literature review to identify existing theories, frameworks, and research limitations regarding digital economy, privacy and security concerns in financial technologies, computer-mediated interpersonal

relationships, and value sensitive design. The goal is to establish a solid theoretical foundation that can be used to guide the empirical studies and data analysis in Phase 2 and the design directions in Phase 3.

Phase 2 will involve three empirical studies. Study 1 is to collect social media data (e.g., forum posts) regarding the varied usages and experiences of digital P2P payments systems. Content analysis will be used to identify patterns and general trends related to using and experiencing digital P2P payments systems in people's daily lives.

Study 2 would be an in-depth interview study to investigate people's perceptions and subjective experiences of how they use digital P2P payments apps and how they understand the values and impacts of such technology for their interpersonal relationships. Up to 30 participants will be recruited through multiple methods such as posting on social media and snowball sampling. In-depth qualitative analysis based on the Grounded Theory [5] will be used to provide a rich narrative of the unique role of digital P2P payment apps in people's increasingly intertwined financial and social lives.

Study 3 would be a large-scale anonymous online survey (up to 500 participants) to measure the impacts of digital P2P payments on people's interpersonal relationships and their cultural expectations for such technology. This survey will be distributed in multiple ways, such as Twitter, Reddit, and Amazon Mechanical Turk. Measures and specific survey questions will be designed based on findings from Study 1 and 2. The goal of this study is to reach a broader audience and improve the generalizability of the findings.

Grounded in the theoretical foundation from Phase 1 and findings from Phase 2, Phase 3 will focus on identifying requirements and creating prototypes for redesigning digital P2P payments services for supporting both secure and private financial transactions and nuanced social interaction and interpersonal connections. Principles of value-sensitive design will be used to highlight the importance of sociocultural values in redesigning financial technologies such as digital P2P payments apps. Standard prototype tools and an iterative research-design-development process will be used to create functional digital prototypes for future user studies. The goal is to design social and cultural sensitive digital P2P payments platforms that can better serve people's needs for conducting technology-mediated financial activities as interactive social experiences and reducing social awkwardness in this process.

5 PRELIMINARY RESULTS

To date I have conducted two preliminary studies to investigate how using digital P2P payments may affect the ways through which people socially engage and connect with each other. The first study focuses on how people perceive the increasing trend of integrating P2P payments with social media services (e.g., Facebook Messenger payment; see Figure 1) and why they decide not to use this service in their social lives. Using 158 social media posts and 8 in-depth interviews, findings highlighted three main sociotechnical barriers that hinder people from using Messenger payment: the established "stickiness" with other digital P2P payment methods, the existing distrust of the Facebook ecosystem, and tensions between financial transaction and personal interaction within the same platform. This

work has been published at HICSS 2021 – The Hawaii International Conference on System Sciences [17].

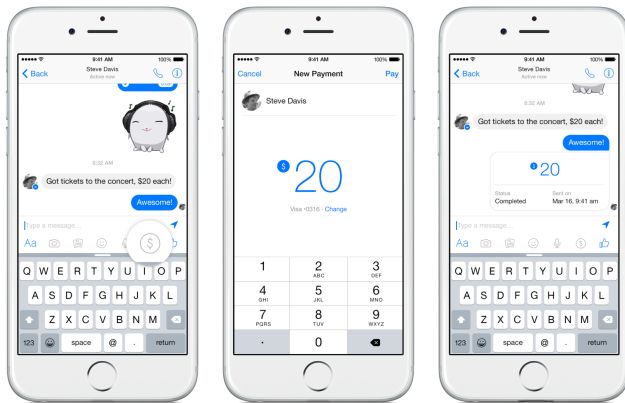


Figure 1: The P2P payment feature in Facebook Messenger (source: Facebook)

The second study investigates the interplay of financial exchanges and offline interpersonal relationships through digital P2P payments based on 23 in-depth interviews. Findings show that using digital P2P payments helps reduce awkwardness and ensure a stronger sense of fairness in financial exchanges. In addition, in everyday social interactions, although digital P2P payments can relieve tension and reduce distrust, they also result in loss of emotion and increase peer pressure. This work has been published by *Telematics and Informatics journal*.

6 CONTRIBUTIONS

My research will contribute to better understanding new and more complicated social phenomena and dynamics emerging in today's digital economy. First, I will offer empirical evidence to unpack and explicate the multidimensional influences of digital P2P payments on both experiences of financial transactions and everyday social interactions, which is understudied in prior scholarship. In doing so, I will provide new perspectives on today's technology-mediated financial life and shed light on the intertwining financial transactions and social relationships through technology. Second, I will design and develop prototypes that highlight the importance of taking the interplay of financial and social engagement, in addition to security and privacy, into consideration when designing digital P2P payments platforms. These design concepts and directions would inform the design of future financial technologies to help people feel more comfortable, interactive, and emotionally satisfactory when using such technologies. In particular, with the concern of the current COVID-19 pandemic, there is an increasing interest to make digital financial technologies not only secure but also more human-centered, interaction-centric, and culture sensitive, which can be used to better support and maintain human connections through daily financial activities without face-to-face interaction. Therefore, in a broader sense, my research on the social values of digital P2P payments also contributes to building a more robust and inclusive digital economy in today's changing society.

7 NEXT STEPS

I am currently collecting data for my dissertation in summer 2021. I also plan to defend my dissertation proposal before October 2021. Therefore, it would be very valuable and beneficial for me to attend the doctoral consortium and receive feedback on my research and future career path from peers and senior researchers. I plan to defend my dissertation and graduate in the fall of 2022.

8 DISSERTATION STATUS AND LONG-TERM GOALS

Currently, I am collecting data and drafting my dissertation proposal. I plan to defend my dissertation proposal before October 2021. I am also applying for PhD fellowships to seek funding to support data collection for my dissertation. For my long term goals, I would like to turn my research into practice by applying my research findings to related industries and help design and develop financial products that also take social engagement and cultural values into account. For example, I would like to help Facebook integrate P2P payments into its ecosystem (e.g., Messenger, WhatsApp, Instagram) in a more seamless and human-centered way that helps users feel more comfortable, interactive, and emotionally satisfactory. In this way, I hope that my research would contribute to building a more robust and organic digital economy in the future.

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